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Editor-in-Chief
IEEE Transactions on Energy Conversion

Dear Editor,

We submit for your consideration the manuscript:

Extending the SAMBPS Estimation Window Beyond 150 ms: A Nine-State VSM Model with Full AVR and PSS Dynamics

Background and motivation. Model-based adaptive protection of synchronous and virtual synchronous machine (VSM) units requires accurate fault-current estimation over the full protection operating interval. Our prior work (under review, *IEEE Transactions on Power Systems*) demonstrated that a sixth-order Park dq truth model limits the self-adaptive model-based protection system (SAMBPS) to estimation windows of at most 150 ms. Beyond this boundary, automatic voltage regulator (AVR) and power system stabiliser (PSS) dynamics introduce unmodelled excitation transients that raise the Levenberg–Marquardt condition number κ_n above the SAMBPS veto threshold, causing false-restrain outcomes for slow-clearing faults and island-detection scenarios.

Contributions. This paper makes four distinct contributions:

1. **Nine-state VSM model (C1):** A stiff ODE formulation coupling six Park dq flux states with one IEEE Type-1 AVR excitation state and two lead-lag PSS states, with a virtual-impedance droop interface.
2. **Observability analysis (C2):** Proof that the nine-state model is locally observable from terminal (V_t, I_t, ω) measurements with $L^* \geq 40$ ms, with explicit conditions on fault current and PSS gain.
3. **Window extension proposition (C3):** Formal proof that incorporating AVR and PSS states extends the $\kappa_n < 30$ operating region from $L^* \leq 150$ ms to $L^* \leq 380$ ms (+230 ms, +153%), subject to $\pm 10\%$ parameter knowledge.
4. **Islanded-microgrid case study (C4):** On a 10-bus benchmark, zone detection accuracy at 300 ms improves from 84.0% (6th-order) to 97.1% (9-state), and all island-detection scenarios that the 6th-order model failed are correctly resolved.

Fit to the journal. *IEEE Transactions on Energy Conversion* is the primary venue for machine modelling, excitation control, and generator dynamics studies. This manuscript directly connects AVR and PSS dynamics to protection system performance — a bridge between the energy conversion and power systems communities that fits squarely within the journal’s scope.

Suggested reviewers.

- Prof. Q.-C. Zhong (Illinois Institute of Technology) — virtual synchronous machine control
- Prof. Marta Molinas (NTNU) — GFM inverter dynamics
- Prof. Dionysios Aliprantis (Purdue University) — machine modelling and simulation

- Dr. Silvio D'Arco (SINTEF Energy Research) — VSM stability and control

This manuscript is not under review elsewhere. All authors have approved the submission.

Sincerely,

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